Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester III)

Scala For Data Science

Practical XIV

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Class:SYBsc IT	Subject: Scala For Data Science
Date of Assignment:11/08/2026	Date/Time of Submission:11/08/2026

AIM: Perform basic time series analysis in Scala. Generate synthetic time series data (e.g., daily sales over a month).

CODE:

build.sbt DEMO.scala

```scala
import scala.util.Random

object Demo1 extends App{

  println("Day\tSales")

  for(day <- 1 to 30){
    val sales = 50 + day * 2 + Random.nextInt(10) -5
    println(s"$day\t$sales")
  }
  println("IQRA S021")
}
```

OUTPUT:

```
4   60
5   63
6   61
7   65
8   62
9   68
10  67
11  68
12  78
13  72
14  79
15  80
16  81
17  86
18  87
19  92
20  89
21  92
22  96
23  96
24  98
25  101
26  98
27  100
28  101
29  105
30  106
IQRA S021

Process finished with exit code 0
```

PRACTICAL XV

AIM: **Create polynomial features from a dataset. Given a list of numbers (e.g., [1, 2, 3]), generate polynomial features up to degree 3.**

CODE:

```scala
object Demo extends App{

  val data = List(2,3,4,5,6)

  for (x <- data){
    val p1 = math.pow(x,1)
    val p2 = math.pow(x,2)
    val p3 = math.pow(x,3)

    println(s"$x\t$p1, \t$p2, \t$p3")
  }
  println("IQRA S021")
}
```

<u>OUTPUT:</u>

```
"C:\Program Files\Java\jdk1.8.0_181\bin\ja
2   2.0,    4.0,    8.0
3   3.0,    9.0,    27.0
4   4.0,    16.0,   64.0
5   5.0,    25.0,   125.0
6   6.0,    36.0,   216.0
IQRA S021

Process finished with exit code 0
```

<u>PRACTICAL XVI</u>

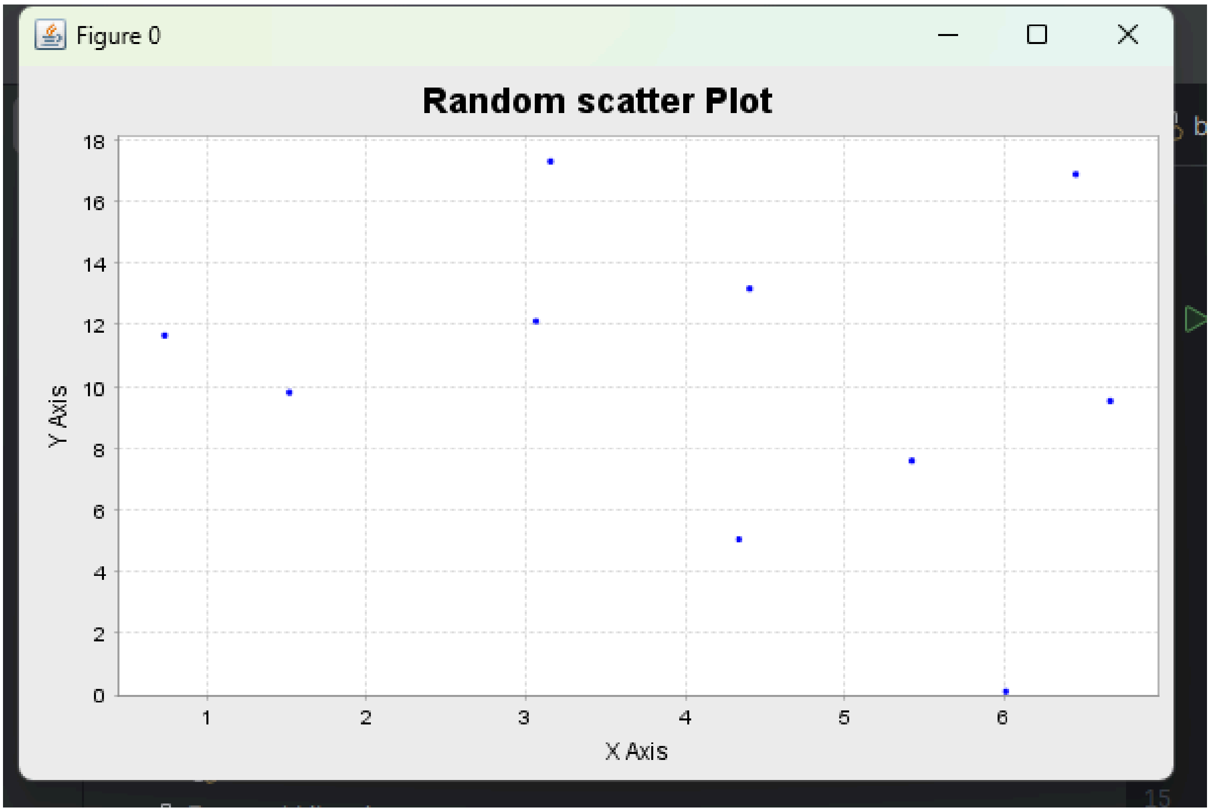
<u>AIM:</u>Create a scatter plot of random data using Breeze-viz. Label the axes and customize the color of points.

<u>CODE:</u>

```scala
import breeze.plot._
import scala.util.Random

object DEMO extends App {
  val x = List.fill(10)(Random.nextDouble() * 10)
  val y = List.fill(10)(Random.nextDouble() * 20)
  println(x)
  println(y)
  val f = Figure()
  val p = f.subplot(0)

  p += plot(x, y, '.', colorcode = "blue")

  p.xlabel = "X Axis"
  p.ylabel = "Y Axis"
  p.title = "Random scatter Plot"
  f.refresh()
  println("IQRA S021")
 }

```

<u>OUTPUT:</u>

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
List(4.404541260829279, 3.0653800370485875, 1.5166901714670855, 3.1550692275777625, 4.337
List(13.16088728744877, 12.109201481182188, 9.806265139473936, 17.28819042817898, 5.03044
IQRA S021
```



PRACTICAL XVII

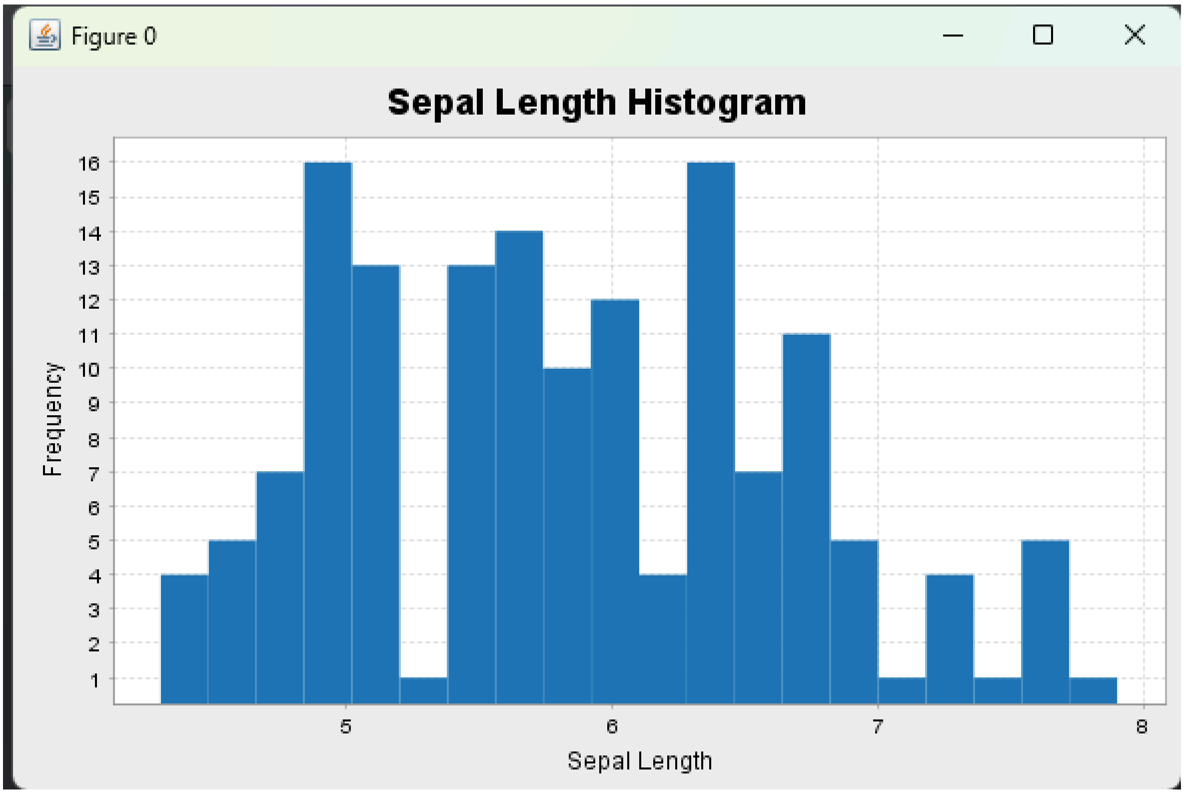
AIM: **Generate a histogram of a dataset using Breeze-viz. Experiment with different bin sizes.**

CODE:

build.sbt DEMO.scala

```scala
import breeze.plot._
import scala.io.Source

object Demo extends App{
      val file = Source.fromResource("iris.csv")
  val data = file.getLines()
    .drop(1)
    .map(line => line.split(",")(0).toDouble)
    .toList
  file.close()
  val fig = Figure()
  val plt = fig.subplot(0)
  plt += hist(data,bins = 20)
  plt.title = "Sepal Length Histogram"
  plt.xlabel = "Sepal Length"
  plt.ylabel = "Frequency"
  fig.refresh()
  println("IQRA S021")
 }
```

OUTPUT:



PRACTICAL XVIII

AIM: Plot a line graph for a dataset showing a trend over time.

CODE:

build.sbt DEMO.scala

```scala
object Demo extends App{
      val file = Source.fromResource("iris.csv")
  val x = file.getLines()
    .drop(1)
    .map(line => line.split(",")(0).toDouble)
    .toList
  file.close()
  val y = x.indices.map(_.toDouble).toList

  val fig = Figure()
  val plt = fig.subplot(0)

  plt += plot(y,x)

  plt.xlabel = "Record Number"
  plt.ylabel = "Sepal Length"
  plt.title = "Sepal Length Trend"
  fig.refresh()
  println("IQRA S021")
}
```

OUTPUT:

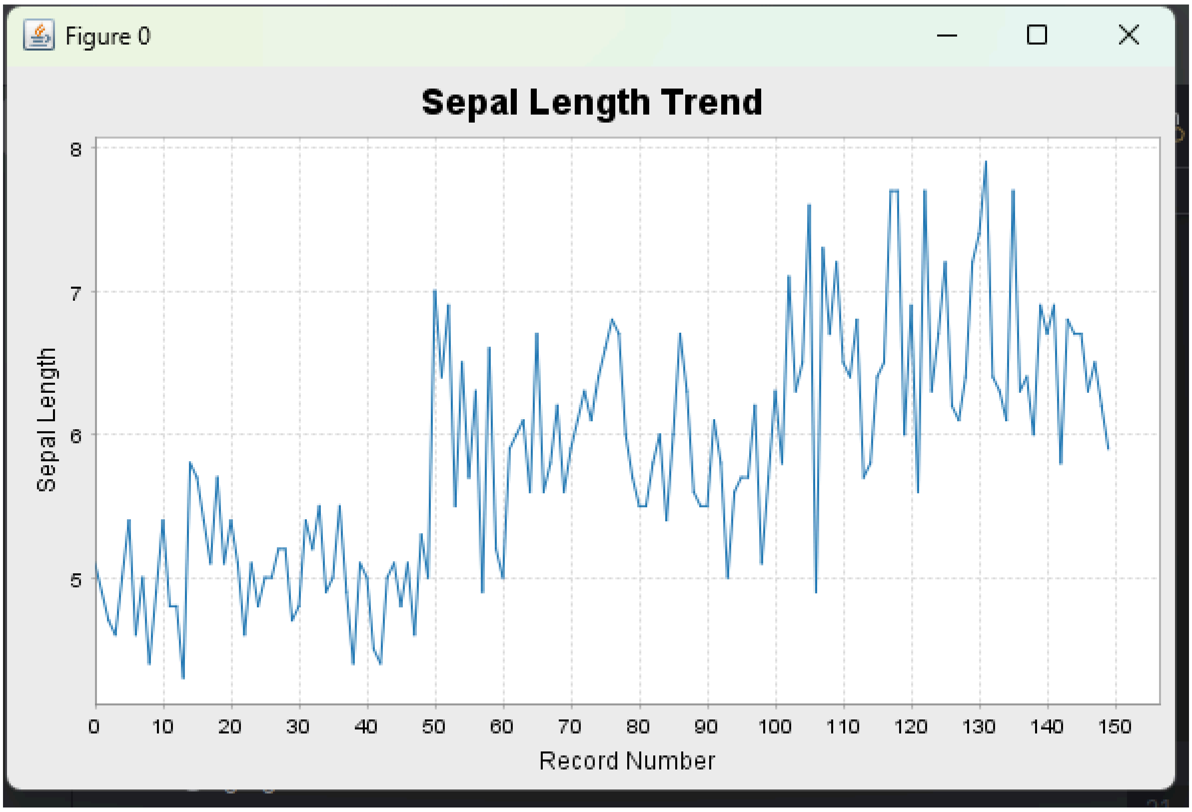


Figure 0
Sepal Length Trend
Sepal Length
Record Number
8
7
6
5
0
10
20
30
40
50
60
70
80
90
100
110
120
130
140
150